

A Real-Time Streaming Bridge Between Audio Encoders and Small Language Models via Adaptive Token Compression

Supriya Adhikari[‡] Kristina Ghimire^{*} Shital Adhikari^{*} Sanjay Rijal^{*}
You Young^{*} Vaghawan Ojha^{*}

Abstract

Recent advances in large language models (LLMs) have enabled strong multimodal reasoning capabilities [7, 5]; however, integrating continuous audio streams with small, efficient LLMs remains a fundamental challenge [2, 5]. Audio encoders such as Whisper, [4] gives dense frame-level representations at high temporal resolutions, which results in prohibitive context lengths and latency when directly consumed by downstream language models. This creates a mismatch between continuous sensory inputs and discrete token-based reasoning systems, particularly in real-time and resource-constrained environments.

Streaming sequence modeling has been extensively studied through monotonic [3] and chunk-wise attention mechanisms, such as MoChA [6], which enforce causal and low-latency processing constraints. More recent systems, including Moshi [1] other real-time speech interaction models, extend these ideas to practical streaming settings. While these approaches focus on controlling attention flow or decoding latency, they do not explicitly address the mismatch between dense audio representations and discrete token-based LLM interfaces (at least to our knowledge). In contrast, our work introduces an adaptive compression layer that transforms continuous audio into a token-efficient representation suitable for streaming interaction with frozen small language models.

In this work, we propose a streaming adapter framework that enables a real-time bridge between audio encoders and small LLMs by compressing continuous audio features into a low-bitrate sequence of semantically meaningful tokens. In our approach, we plan to transform high-frequency acoustic representations into a compact, append-only token stream (1–3 tokens per second), allowing a frozen small LLM to perform incremental reasoning and generation under strict latency constraints.

The core of our method is a trainable adapter network inspired by cross-modal resampling architectures. Given overlapping audio windows, the adapter employs a set of learnable query vectors that attend over encoder-produced frame features, compressing each temporal segment into a small number of tokens. These tokens are appended to the LLM context in a KV-cache-efficient manner, enabling incremental decoding without recomputation. To maintain temporal coherence, we introduce a stability buffer mechanism that smooths token representations across adjacent windows.

A key contribution of our work will be framing the problem through an information bottleneck perspective. The adapter is trained to minimize redundancy while preserving task-relevant semantics, effectively achieving high compression ratios (up to 25:1). We further introduce an adaptive token rate controller that dynamically adjusts the number of tokens per window based on input complexity, allocating more capacity to ambiguous or noisy segments.

To support streaming inference, we propose an early-commit gating mechanism that learns when sufficient information has been accumulated to initiate generation. This explicitly models the latency–accuracy trade-off, enabling early responses while minimizing revision. Additionally,

^{*}E.K. Solutions Pvt. Ltd., Nepal

[‡]The first two authors contributed equally to this work

we define a prefix consistency objective that enforces stability in model outputs as new tokens are appended, reducing undesirable output fluctuations.

Training is conducted in three stages: (1) audio-text alignment via contrastive learning, (2) content preservation through ASR distillation, and (3) task-level distillation from a larger teacher LLM. Importantly, only the adapter (and optional lightweight LLM adapters) is trained, while both the audio encoder and LLM remain frozen, ensuring efficiency and modularity.

We plan to evaluate our approach across three axes: task performance, latency, and stability.

Our framework provides a principled solution for integrating continuous audio signals with discrete reasoning systems, bridging the multimodal learning. We hope this work enables efficient, low-latency speech-driven reasoning with small LLMs and opens new directions for real-time multimodal intelligence in resource-constrained environments.

Architecture

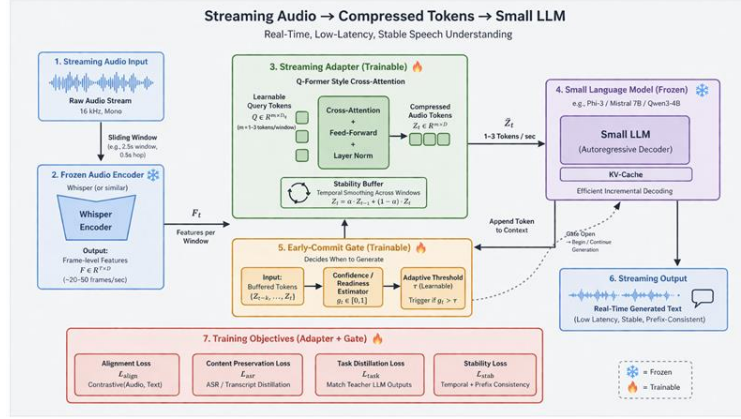


Figure 1: Overview of the Model architecture. Figure Generated using ChatGPT.

Method

Audio input is segmented into overlapping windows and processed by a frozen encoder to produce frame-level features $F \in \mathbb{R}^{T \times D}$. A trainable adapter then compresses these features into a small set of tokens using cross-attention, defined as $Z = \text{Attn}(Q, F)$, where $Q \in \mathbb{R}^{m \times D_q}$ are learnable query vectors. We dynamically control the token generation rate as $R = \frac{m}{t}$ while preserving semantic information through an information bottleneck objective $\max I(Z; Y)$ s.t. $I(X; Z) \leq \beta$. To ensure temporal consistency in the streaming setting, we introduce a stability objective $\mathcal{L}_{\text{stability}} = \sum_t \|Z_t - Z_{t-1}\|^2$, along with a prefix consistency constraint enforcing $P(y | Z_{1:t}) \approx P(y | Z_{1:t+k})$. Finally, an early-commit gating mechanism determines when to initiate generation using $g_t = \sigma(WZ_t)$, explicitly balancing the trade-off between latency and accuracy.

Notes

We have just begun working on the preliminary experiment setup, and we do not have the results yet, but by the time the summer school starts, we will have some preliminary results. Supriya A. and Kristina G. are the primary contributors of the paper, including methodologies, literature review, all experiments, writing and editing. Supriya A. designed and did all initial experiments across all tasks and drafted the paper.

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